Physics 4B Spring 2026
Week 2 Discussion

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1. **Brainwarmer**[1]. Consider the wave $\psi(x,t) = 0.5\cos(100\pi x - 60\pi t)$ on a string. What is the (i) amplitude, (ii) frequency, (iii) maximum velocity of a point, and (iv) maximum acceleration?

   Solution:

   (i) The amplitude is 0.5.

   (ii) The frequency is $f = \frac{60\pi}{2\pi} = 30$.

   (iii) The velocity is given by $v(x,t) = \dot{\psi}(x,t) = 30\pi\sin(100\pi x - 60\pi t)$, so the maximum velocity is 30π.

   (iv) The acceleration is given by $a(x,t) = \ddot{\psi}(x,t) = -1800\pi^2\cos(100\pi x - 60\pi t)$, so the maximum velocity is $1800\pi^2$.

2. **More Driven Oscillator**. A 0.5 kg mass is connected to a spring with $k = 200$ N/m. The system is immersed in a fluid providing a damping coefficient $\gamma = 0.2$ s^{-1}. It is driven by an external force $F(t) = 5\cos(18t)$ N.

   (a) Find the resonant frequency.

   (b) Find the amplitude at the resonant frequency.

   Solution:

   (a) Let's start with the solution that we derived last time (in complex form):

   $$z(t) = \frac{F_0}{mZ(\omega)}e^{i\omega t} \tag{1}$$

   where $Z(\omega) = -\omega^2 + i\gamma\omega + \omega_0^2$.

   To find the resonant frequency, we can try to write $z(t)$ in polar form so that the amplitude is isolated out. We start by writing $Z(\omega) = |Z|e^{i\omega t}$ which leads to

   $$|Z| = \sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \tag{2}$$

   Since we are writing $z(t)$ in polar form, the amplitude for $x(t) = \text{Re}z(t)$ is

   $$A = \frac{F_0}{m|Z|} = \frac{F_0}{m\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}} \tag{3}$$

[1] Shankar 9.5

The resonant frequency is the value of ω that maximizes A, which happens when $|Z|$ is minimized:

$$\frac{d|Z|}{d\omega} = \frac{\omega(\gamma^2 - 2(\omega_0^2 - \omega^2))}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}} = 0 \tag{4}$$

This has three solutions $\omega = 0, \pm\sqrt{\omega_0^2 - \gamma^2/2}$. You can check that the first one is a local maximum, while the other two are minima, so the resonant frequency is $\omega = \sqrt{\omega_0^2 - \gamma^2/2}$.

After plugging in the numbers, we get $\omega \approx 20$ rad/s.

(b) Plug the number for ω from (a) into (3), using $F_0 = 5$ N and $\gamma = b/m = 0.2$ s^{-1}, we get $A = 2.5$ m.

3. **Traveling Wave**. A mass M hangs from a rope of length L and mass $m \ll M$. We give the lower end a transverse flick.

(a) How long will it take to reach the top? Use the following numbers: $L = 8$ m, $m = 50$ g, $M = 5$ kg.

(b) Assume the pulse we created takes the form $\psi(x,t) = Ae^{-B(x-vt)^2}$, where $A = 0.02$ m and $B = 2$ m^{-2}. Find the instantaneous power transmitted by the rope at $x = 1$ m and $t = 0$ s.

Solution:

(a) Since we assume the mass M is much more massive than the rope, we can ignore the weight of the rope. The tension along the rope is therefore $T = Mg$. The time it takes the pulse to travel the distance L is

$$t = \frac{L}{v} = L\sqrt{\frac{\mu}{T}} = L\sqrt{\frac{m}{L \cdot Mg}} = \sqrt{\frac{mL}{Mg}} = 0.09 \text{ s}$$

(b) The kinetic energy per unit length along the rope is

$$\frac{dK}{dx} = \frac{1}{2}\mu\dot{\psi}^2 = 2\mu A^2 B^2 v^2 (x - vt)^2 e^{-2B(x-vt)^2} \tag{5}$$

The potential energy per unit length along the rope is

$$\frac{dU}{dx} = \frac{1}{2}T\left(\frac{d\psi}{dx}\right)^2 = 2TA^2B^2(x - vt)^2 e^{-2B(x-vt)^2} = \frac{dK}{dx} \tag{6}$$

The total energy per unit length is therefore

$$\frac{dE}{dx} = \frac{dK}{dx} + \frac{dU}{dx} = 4TA^2B^2(x - vt)^2 e^{-2B(x-vt)^2} \tag{7}$$

and the instantaneous power transmitted is

$$P = \frac{dE}{dt} = \frac{dE}{dx}\frac{dx}{dt} = \frac{dE}{dx}v = 4TA^2B^2v(x-vt)^2e^{-2B(x-vt)^2} \tag{8}$$

At $x = 1$ m and $t = 0$ s, this is $P \approx 0.5$ W.

4. **Standing Wave**. A high-tension wire is fixed at both ends ($x = 0$ and $x = L$). The wire has a linear mass density $\mu = 5 \times 10^{-4}$ kg/m and is under a tension of $T = 800$ N. The wire is driven such that it vibrates in its fourth harmonic ($n = 4$) at a frequency of $f = 800$ Hz.

   (a) Calculate the phase velocity v.

   (b) Identify the number of nodes and the number of anti-nodes (including the ends).

   (c) Based on the frequency and the harmonic number, determine the total length L of the string.

   (d) The maximum transverse displacement of the standing wave is $\psi_{\max} = 3$ cm. Write the complete mathematical expression for the standing wave $\psi(x,t)$.

   (e) Calculate the total energy stored in the string while it is vibrating in this mode.

   Solution:

   (a) The wave velocity is $v = \sqrt{T/\mu} \approx 1265$ m/s.

   (b) The string is fixed on both ends, so the first harmonic has 2 nodes and 1 anti-node. Every time n goes up by 1, the standing wave gets one more node and one more anti-node, so there are 5 nodes and 4 anti-nodes.

   (c) Given the boundary condition, we know the length is equal to 2 wavelengths since there is a half wavelength between each two adjacent nodes

$$L = 2\lambda = \frac{2v}{f} = 3.16 \text{ m/s} \tag{9}$$

   (d) The amplitude is equal to the maximum displacement, and occurs at the anti-nodes ($x = L/8, 3L/8, 5L/8, 7L/8$). The boundary condition requires that $\psi(0,t) = \psi(L,t) = 0$, so this implies the spatial part has to be $\sin(kx)$, where $k = 2\pi/\lambda = 4\pi/L \approx 3.97$ rad/m. The angular frequency is $\omega = vk = \frac{4\pi}{L}\sqrt{\frac{T}{\mu}} \approx 5026.5$ rad/s. Therefore, the complete expression for the standing wave is

$$\psi(x,t) = A\sin(kx)\cos(\omega t) = 0.03 \text{ m}\sin(3.97 \text{ rad/s}\cdot x)\cos(5026.5 \text{ rad/s}\cdot t) \tag{10}$$

(e) To find the total energy, we first consider the kinetic energy. The transverse velocity of the wave is

$$\dot{\psi}(x,t) = -\omega A \sin(kx)\sin(\omega t) \tag{11}$$

so the total kinetic energy averaged over a period Δt is

$$\begin{aligned}\langle \mathrm{KE}\rangle &= \frac{1}{\Delta t}\int_0^{\Delta t} dt \int_0^L dx\ \frac{1}{2}\mu\dot{\psi}(x,t)^2 \\ &= \frac{1}{2\Delta t}\mu\omega^2 A^2 \int_0^{\Delta t} dt \int_0^L dx \sin^2\left(\frac{4\pi x}{L}\right)\sin^2\left(\frac{2\pi t}{\Delta t}\right) \\ &= \frac{1}{8}\mu\omega^2 A^2 L\end{aligned} \tag{12}$$

The potential energy comes from the stretching of the string segments due to the string tension, so

$$\begin{aligned}\langle U\rangle &= \frac{1}{\Delta t}\int_0^{\Delta t} dt \int_0^L dx\ T\left[\sqrt{1+\left(\frac{d\psi}{dx}\right)^2}-1\right] \\ &\approx \frac{1}{2\Delta t}T\int_0^{\Delta t} dt \int_0^L dx\ \left(\frac{d\psi}{dx}\right)^2 \\ &= \frac{1}{2\Delta t}Tk^2A^2 \int_0^{\Delta t} dt \int_0^L dx\ \cos^2\left(\frac{4\pi x}{L}\right)\cos^2\left(\frac{2\pi t}{\Delta t}\right) \\ &= \frac{1}{8}Tk^2A^2L = \frac{1}{8}\mu\omega^2A^2L\end{aligned} \tag{13}$$

Putting everything together, we get

$$\langle E\rangle = \frac{1}{4}\mu\omega^2A^2L \approx 8.99\ \mathrm{J} \tag{14}$$

5. **More Standing Wave**. A string is anchored on a wall at one end while the other end goes over a pulley and supports a weight $M_1 = 1$ kg. The string is plucked to create a standing wave, which has 3 nodes (including at the ends). When the weight M_1 is replaced by another weight M_2, the number of nodes (including the ends) becomes 5, and the frequency is the same as before.

   (a) What is M_2?

   (b) If the string weighs $m = 0.5$ g and $f = 880$ Hz, what is the length of the string between the wall and the pulley?

   (c) If the maximum amplitude of displacement of points on the string is 1 mm, what is the maximum acceleration of points on the string?

Solution:

(a) For a string fixed on both ends, the frequencies of the harmonic modes are given by

$$f_n = \frac{nv}{2L} = \frac{n}{2L}\sqrt{\frac{T}{\mu}} = \frac{n}{2L}\sqrt{\frac{MgL}{m}} = \frac{n}{2}\sqrt{\frac{Mg}{mL}} \tag{15}$$

3 nodes corresponds to $n = 2$. 5 nodes corresponds to $n = 4$. Since f, g, m and L all stay fixed, we have

$$M_2 = \frac{n_1^2}{n_2^2} M_2 = \frac{1}{4} M_1 = 0.25 \text{ kg} \tag{16}$$

(b) The length is

$$L = \frac{n^2 Mg}{4 f_n^2 m} \approx 0.025 \text{ m} \tag{17}$$

(c) Given the boundary conditions, the standing wave takes the form

$$\psi(x,t) = A \sin(kx) \cos(\omega t) \tag{18}$$

with $k = n\pi/L$ and $\omega = vk$. The acceleration would look like

$$\ddot{\psi}(x,t) = -\omega^2 A \sin(kx) \cos(\omega t) \tag{19}$$

with its maximum being

$$\ddot{\psi}_{\max} = \omega^2 A = 4\pi^2 f A \approx 3 \times 10^4 \text{ m/s}^2 \tag{20}$$